Theoretical Knowledge

Incident Escalation Workflows

Incident escalation workflows define how security incidents are managed and escalated within a Security Operations Center. Because security teams handle a large number of alerts daily, structured workflows are necessary to ensure that high-risk incidents are properly investigated and handled by the appropriate personnel. Escalation processes allow analysts to prioritize threats, coordinate investigations, and communicate effectively with other teams and stakeholders.

Most SOC environments operate using a **tiered structure**. Tier 1 analysts are responsible for alert triage and initial investigation. They review alerts generated by security tools, determine whether the alert is a false positive or a legitimate threat, and collect preliminary evidence. If the alert indicates a serious or complex issue, it is escalated to Tier 2 analysts.

Tier 2 analysts perform deeper investigation and analysis of confirmed incidents. They examine logs, analyze network traffic, review endpoint activity, and determine the scope and impact of the incident. If the incident requires advanced expertise or specialized investigation techniques, it may be escalated further to Tier 3 analysts.

Tier 3 analysts are typically senior security specialists who perform advanced threat hunting, malware analysis, and forensic investigations. They may also develop detection rules and improve security monitoring systems based on lessons learned from previous incidents.

Effective incident escalation also requires clear **communication protocols**. One commonly used communication format is the **Situation Report (SITREP)**, which summarizes key information about an incident. A SITREP typically includes the incident description, timeline of events, affected systems, actions taken, and recommended next steps. These reports help ensure that all stakeholders, including management and technical teams, have a clear understanding of the situation.

Modern SOC environments often use **Security Orchestration, Automation, and Response (SOAR)** platforms to automate parts of the escalation process. SOAR tools can automatically assign incidents to appropriate analysts, enrich alerts with threat intelligence, and trigger automated response actions. Automation helps reduce response times and allows analysts to focus on more complex investigation tasks.

The primary objective of incident escalation workflows is to ensure that security incidents are handled efficiently, accurately, and consistently. A well-designed escalation process allows SOC teams to respond quickly to high-priority threats while maintaining effective communication between analysts and stakeholders.

To learn incident escalation workflows, analysts can study incident response guidelines such as the NIST SP 800-61 Computer Security Incident Handling Guide. The SANS Incident Handler’s Handbook also provides practical examples of incident management processes and communication templates used in real-world SOC environments. Additionally, studying SOAR

platforms such as Splunk SOAR helps analysts understand how automation can streamline incident management and escalation processes.